



Cardinality, countable sets

Problem 7 Show that the following sets are countable.

- (a) The set of negative integers.
- (b) The set of non-negative integers that are divisible by 3.
- (c) The set $\mathbb{Q}^2$.

Solution.

- (a) The set of negative integers can be listed as

$$\{-1, -2, -3, -4, \dots\}.$$

- (b) The set of non-negative integers that are divisible by 3 can be listed as

$$\{0, 3, 6, 9, 12, \dots\}.$$

- (c) We already know that $\mathbb{Q}$ is countable. We list it as a sequence

$$\mathbb{Q} = \{q_1, q_2, q_3, q_4, \dots\}.$$

The set $\mathbb{Q}^2$ consists of pairs (q_i, q_j) , $i, j \in \mathbb{N}$. It can be listed as a sequence by applying the same idea as in the proof of the fact that $\mathbb{Q}$ is countable — we write the pairs in a infinite square table and list them by going along the diagonals:

(q_1, q_1)	(q_1, q_2)	(q_1, q_3)	(q_1, q_4)	$\dots$
	$\swarrow$		$\swarrow$	
(q_2, q_1)	(q_2, q_2)	(q_2, q_3)	(q_2, q_4)	$\dots$
	$\swarrow$		$\swarrow$	
(q_3, q_1)	(q_3, q_2)	(q_3, q_3)	(q_3, q_4)	$\dots$
	$\swarrow$		$\swarrow$	
(q_4, q_1)	(q_4, q_2)	(q_4, q_3)	(q_4, q_4)	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Now

$$\mathbb{Q}^2 = \{(q_1, q_1), (q_1, q_2), (q_2, q_1), (q_1, q_3), (q_2, q_2), (q_3, q_1), (q_1, q_4), \dots\}.$$

Problem 8 Prove or disprove:

- (a) The union of three countable sets is countable.
- (b) The intersection of three countable sets is countable.

Solution.

- (a) True. Let the three sets be

$$A = \{a_1, a_2, a_3, a_4, \dots\},$$

$$B = \{b_1, b_2, b_3, b_4, \dots\},$$

$$C = \{c_1, c_2, c_3, c_4, \dots\}.$$

We list

$$A \cup B \cup C = \{a_1, b_1, c_1, a_2, b_2, c_2, a_3, b_3, c_3, \dots\},$$

and remove the repeats.

- (b) False. The intersection of three countable sets can be finite or even empty. For example, if

$$A = \{3k - 2 : k \in \mathbb{N}\} = \{1, 4, 7, 10, 13, \dots\},$$

$$B = \{3k - 1 : k \in \mathbb{N}\} = \{2, 5, 8, 11, 14, \dots\},$$

$$C = \{3k : k \in \mathbb{N}\} = \{3, 6, 9, 12, 15, \dots\},$$

then

$$A \cap B \cap C = \emptyset.$$

Problem 9 Let $a, b \in \mathbb{R}$, $a < b$. Show that the intervals $(0, 1)$ and (a, b) have the same cardinality. (Hint: Construct a linear bijection.)

Solution.

Take the bijection $f : (0, 1) \rightarrow (a, b)$, $f(t) = a + t(b - a)$.

Limits of sequences

Problem 10 (a) Prove by definition that the sequence $\left(\frac{1}{\sqrt{n}}\right)_{n=1}^{\infty}$ converges to the limit 0.

- (b) Prove by definition that the sequence $(0, 1, 2, 0, 1, 2, 0, 1, 2, \dots)$ diverges.

Solution.

(a) Let $\varepsilon > 0$ be an arbitrary positive number. We have

$$\left| \frac{1}{\sqrt{n}} - 0 \right| = \frac{1}{\sqrt{n}},$$

and we want that $\frac{1}{\sqrt{n}} < \varepsilon$. This inequality is equivalent to $n > \frac{1}{\varepsilon^2}$. By the Archimedean Property, there is $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon^2}$. For all $n \geq N$ we have

$$\left| \frac{1}{\sqrt{n}} - 0 \right| = \frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \varepsilon.$$

Thus, by definition $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$.

(b) Let $(a_n)_{n=1}^{\infty} = (0, 1, 2, 0, 1, 2, 0, 1, 2, \dots)$. For any $a \in \mathbb{R}$ at least one of the terms $|a - 0|$, $|a - 1|$, $|a - 2|$ exceeds $\frac{1}{2}$. Take $\varepsilon = \frac{1}{2}$. By above, there is no $N \in \mathbb{N}$ such that $|a - a_n| < \frac{1}{2}$ for all $n \geq N$. Thus, by definition, $(a_n)_{n=1}^{\infty}$ diverges.

Problem 11 For the following sequences $(a_n)_{n=1}^{\infty}$, determine the number $N \in \mathbb{N}$ with the property that

$$|a_n - a| < \varepsilon \quad \text{for all } n \geq N,$$

if $\varepsilon = \frac{1}{10}$ and if $\varepsilon = \frac{1}{100}$:

$$(a) \ a_n = \frac{n+2}{n}, \quad (b) \ a_n = \left(\frac{1}{2}\right)^n.$$

Solution.

(a) We have $a = \lim_{n \rightarrow \infty} \frac{n+2}{n} = 1$, so that

$$|a_n - a| = \left| \frac{n+2}{n} - 1 \right| = \frac{2}{n}.$$

First consider $\varepsilon = \frac{1}{10}$. The inequality $\frac{2}{n} < \frac{1}{10}$ holds if and only if $n > 20$. So we take $N = 21$.

Similarly, $\frac{2}{n} < \frac{1}{100}$ if and only if $n > 200$. So we take $N = 201$ for $\varepsilon = \frac{1}{100}$.

(b) In this case $a = \lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n = 0$, and we have

$$|a_n - a| = \left(\frac{1}{2}\right)^n.$$

For $\varepsilon > 0$, the inequality $\left(\frac{1}{2}\right)^n < \varepsilon$ is equivalent to $n > -\frac{\ln \varepsilon}{\ln 2} = \frac{1}{\ln 2} \ln \left(\frac{1}{\varepsilon}\right)$.

For $\varepsilon = \frac{1}{10}$ we have $n > \frac{\ln 10}{\ln 2} = 3.32\dots$, and we take $N = 4$.

For $\varepsilon = \frac{1}{100}$ we have $n > \frac{\ln 100}{\ln 2} = 6.64\dots$, and we take $N = 7$.

Problem 12 Prove that $\lim_{n \rightarrow \infty} a_n = a$ if and only if $\lim_{n \rightarrow \infty} (a_n - a) = 0$.

Solution.

We have the following chain of equivalencies:

$$\begin{aligned} & \lim_{n \rightarrow \infty} a_n = a \\ \iff & \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |a_n - a| < \varepsilon \\ \iff & \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |(a_n - a) - 0| < \varepsilon \\ \iff & \lim_{n \rightarrow \infty} (a_n - a) = 0. \end{aligned}$$